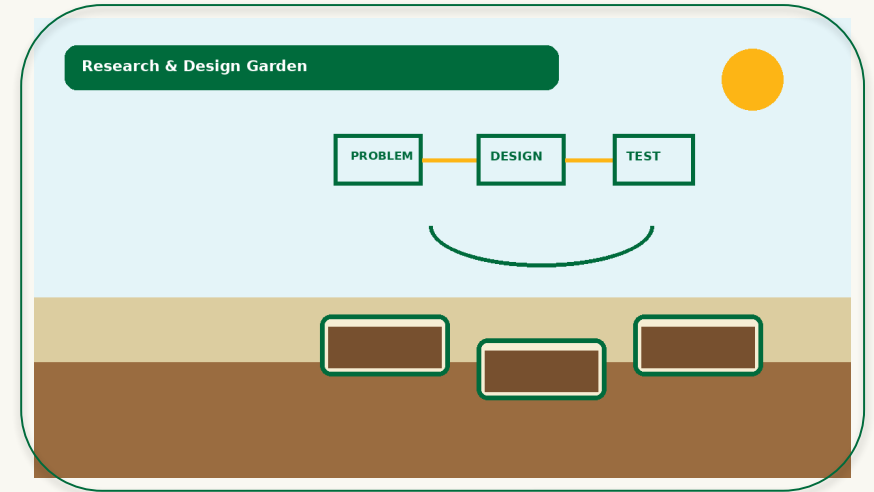


# Research & Design Garden

## Agricultural Engineering Design

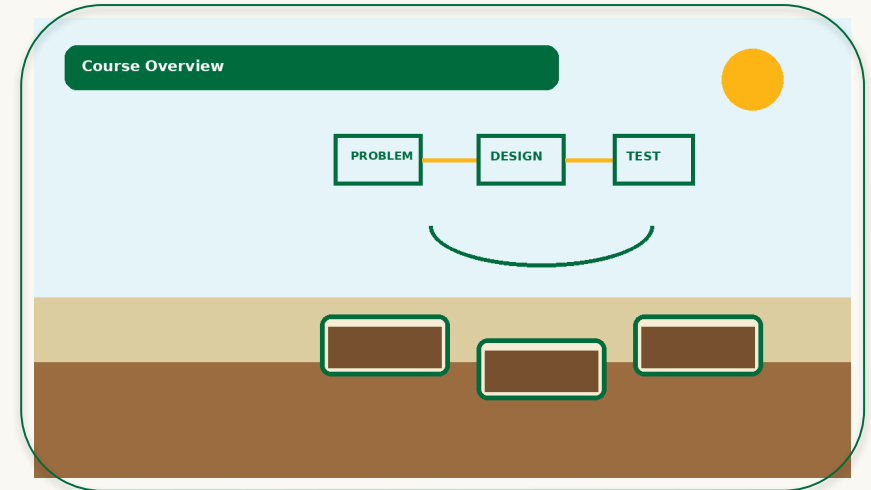
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



# Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide quantitative analysis; distinguish measurement from inference; connect evidence to management, engineering and research decisions.
- Core materials: teacher guide, system drawings, approved tools, low-voltage educational hardware where relevant, measuring instruments, meters/sensors, data sheets/logger, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit an engineering design, monitoring plan, experimental protocol or systems-analysis report appropriate to the course.



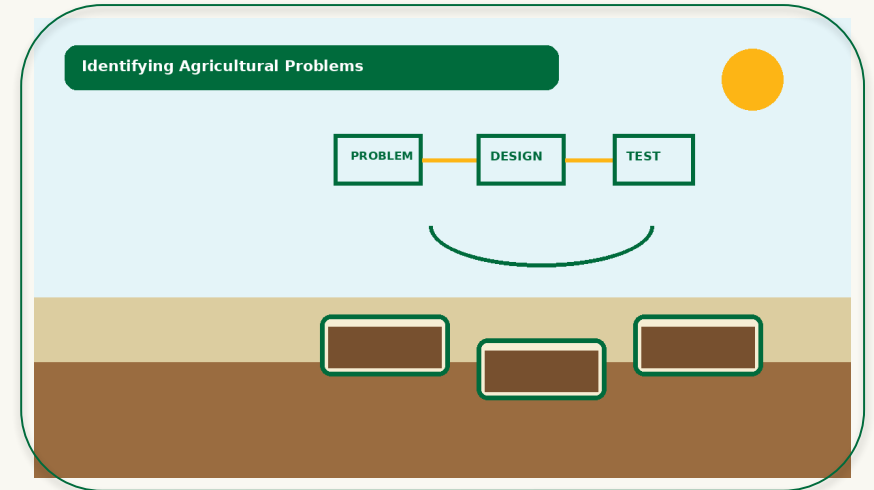
## TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

# Identifying Agricultural Problems

Teacher technical knowledge, laboratory/field practice and student supervision

- Begin with a real production, water, soil, monitoring or postharvest problem observable in the school/community context.
- Separate symptoms from the underlying problem.
- Gather stakeholder/user observations where appropriate.
- Define why the problem matters and who is affected.
- Avoid selecting a solution before the problem is understood.



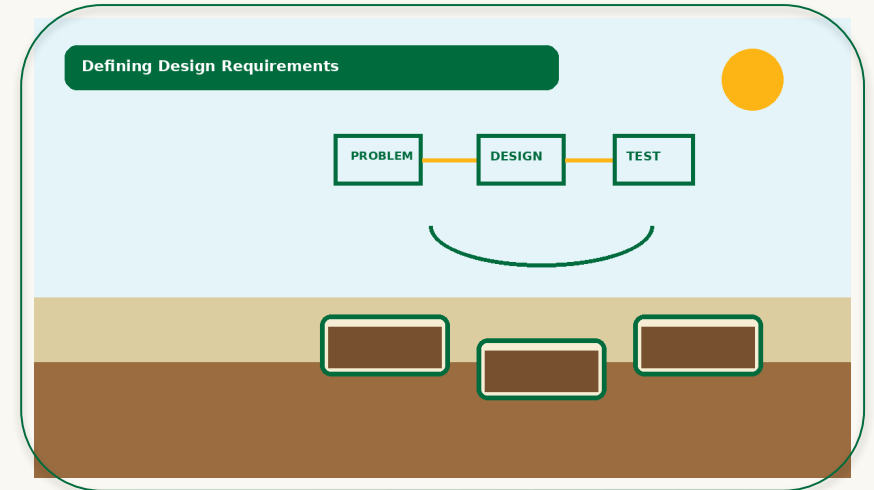
## TEACHER PRACTICE

- Teacher task: facilitate a structured problem-identification session.

# Defining Design Requirements

Teacher technical knowledge, laboratory/field practice and student supervision

- Translate the problem into measurable functional requirements and constraints.
- Include performance, safety, cost, materials, maintenance, size and user requirements.
- Distinguish 'must have' requirements from desirable features.
- Set acceptance criteria before prototyping.
- Avoid requirements that cannot be tested.



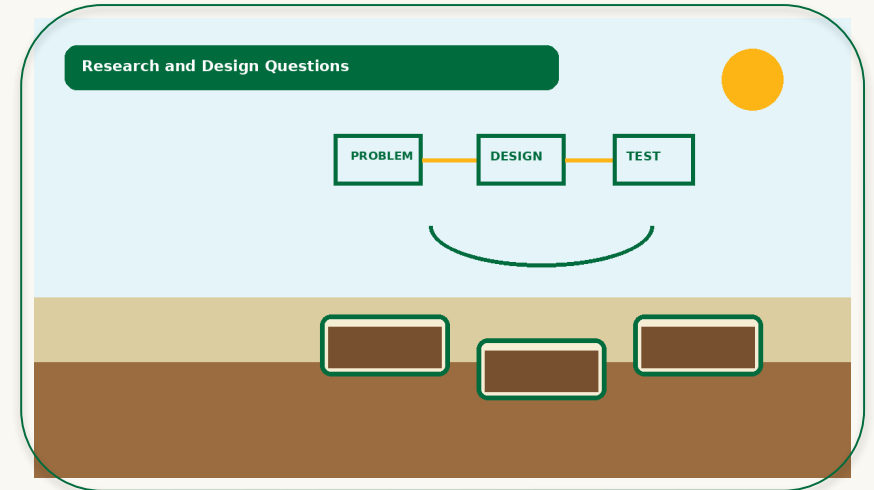
## TEACHER PRACTICE

- Create a design-requirements specification.

# Research and Design Questions

Teacher technical knowledge, laboratory/field practice and student supervision

- Use research questions to understand the problem and design questions to guide solution development.
- Identify missing technical information that must be measured or researched.
- Frame questions around users, environment, performance and constraints.
- Prioritize questions that materially affect design choices.
- Document assumptions that require validation.



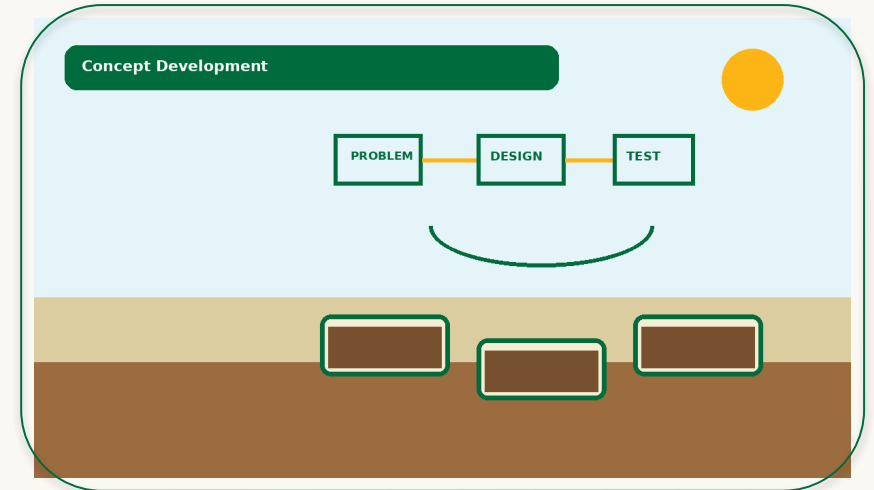
## TEACHER PRACTICE

- Prepare a research-and-design question matrix.

# Concept Development

Teacher technical knowledge, laboratory/field practice and student supervision

- Generate multiple solution concepts before selecting one.
- Use sketches, functional diagrams and morphological combinations where appropriate.
- Compare concepts against requirements rather than personal preference.
- Identify major risks and unknowns.
- Document why a concept advances to prototyping.



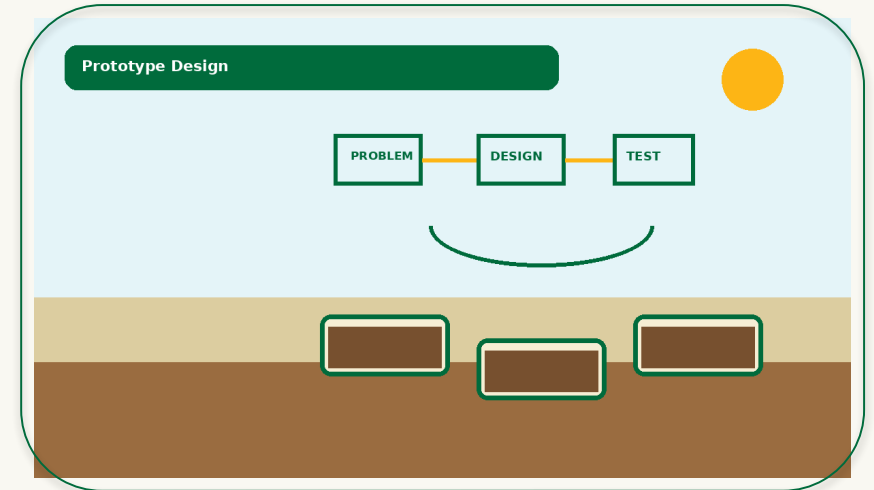
## TEACHER PRACTICE

- Run a three-concept selection exercise.

# Prototype Design

Teacher technical knowledge, laboratory/field practice and student supervision

- Define prototype purpose: proof of concept, subsystem test or integrated demonstration.
- Prepare dimensions, materials, component list and system diagram.
- Use school-safe, low-voltage and structurally stable designs.
- Plan how the prototype will be adjusted or instrumented.
- Obtain adult approval before fabrication.



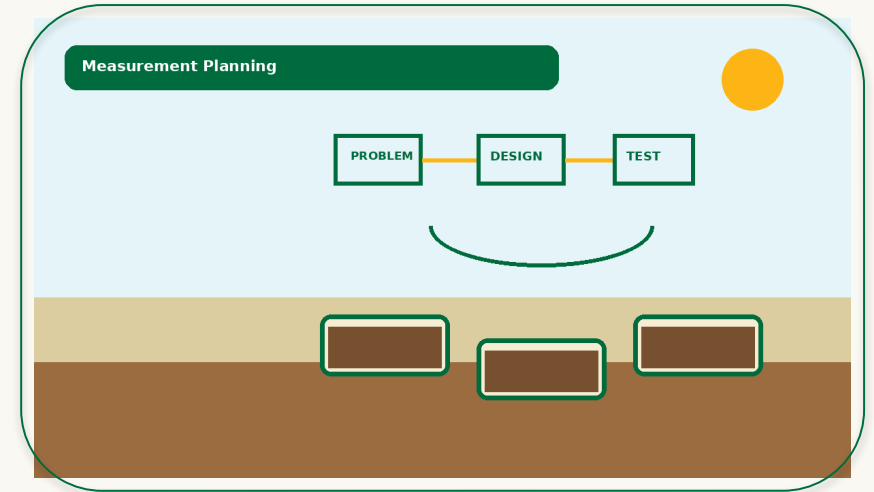
## TEACHER PRACTICE

- Produce a prototype design package.

# Measurement Planning

Teacher technical knowledge, laboratory/field practice and student supervision

- Define performance metrics directly from design requirements.
- Select instruments, units, sampling frequency and test conditions.
- Establish baseline/reference measurements where useful.
- Plan data sheets before testing.
- Include uncertainty and repeat measurements where feasible.



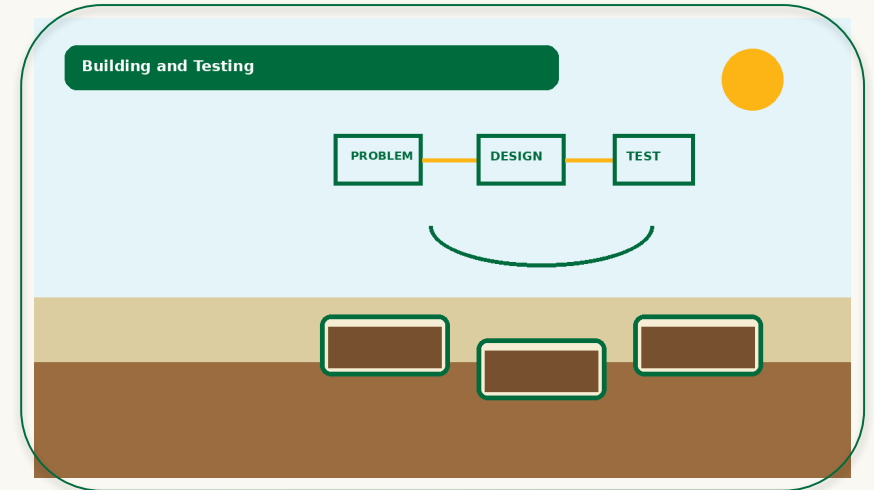
## TEACHER PRACTICE

- Create a prototype test-and-measurement plan.

# Building and Testing

Teacher technical knowledge, laboratory/field practice and student supervision

- Follow drawings, safe tool procedures and adult-only restrictions for hazardous fabrication.
- Inspect the prototype before energizing or loading it.
- Test subsystems before full-system operation.
- Change one design variable at a time when diagnosing performance.
- Record build changes and test conditions.



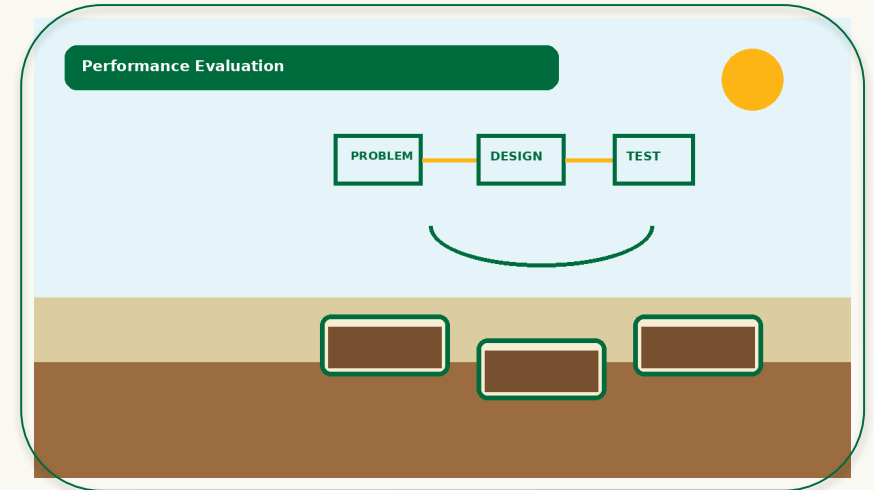
## TEACHER PRACTICE

- Prepare a safe build-test sequence.

# Performance Evaluation

Teacher technical knowledge, laboratory/field practice and student supervision

- Compare measured performance against predefined requirements.
- Use quantitative results plus reliability, usability and maintenance observations.
- Identify requirements met, partially met or unmet.
- Avoid changing success criteria after seeing results.
- Document failure modes as useful engineering evidence.



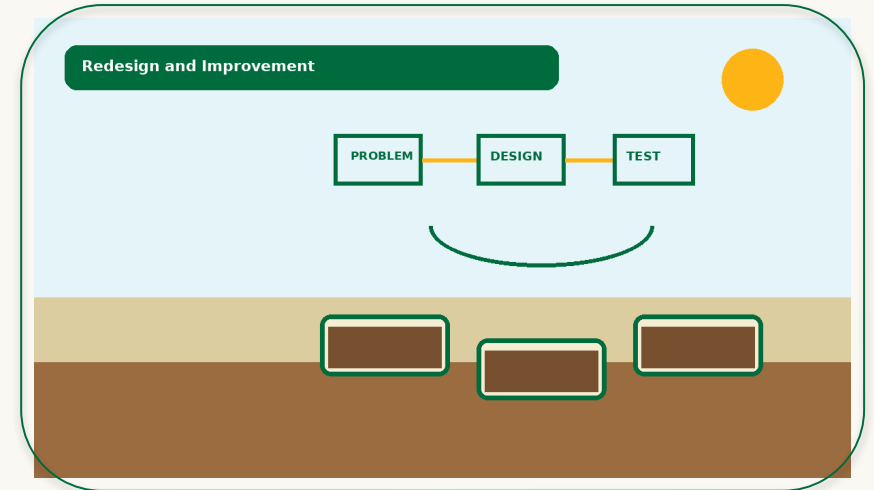
## TEACHER PRACTICE

- Create a requirements-versus-results evaluation table.

# Redesign and Improvement

Teacher technical knowledge, laboratory/field practice and student supervision

- Use test evidence to identify the highest-priority design changes.
- Modify dimensions, components or operating logic with a stated rationale.
- Retest using the same performance metrics.
- Compare version 1 and version 2 results.
- Recognize redesign as a normal part of engineering, not evidence of failure.



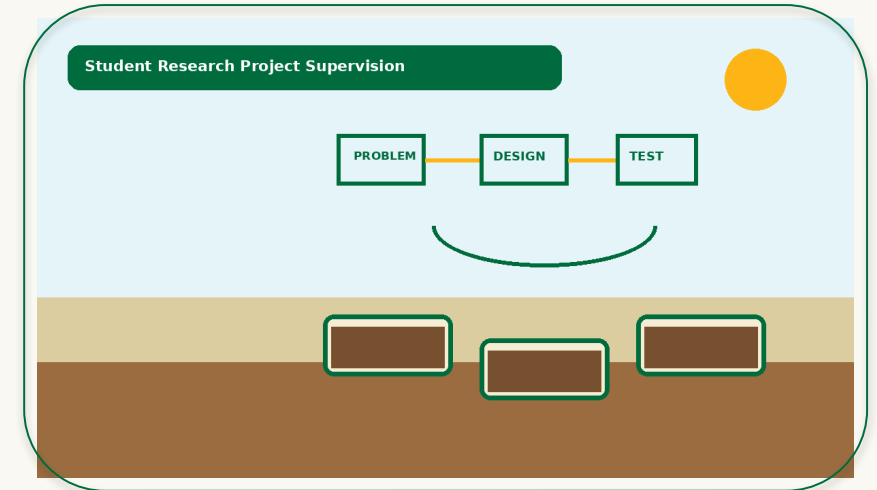
## TEACHER PRACTICE

- Plan one evidence-driven redesign cycle.

# Student Research Project Supervision

Teacher technical knowledge, laboratory/field practice and student supervision

- Require a clear problem, question, method/design, safety review, data plan, analysis and final report.
- Set milestone reviews from proposal through testing.
- Protect research integrity: preserve raw data and report unsuccessful results.
- Ensure projects remain feasible within school resources and time.
- Assess reasoning and evidence, not only the final prototype.



## TEACHER PRACTICE

- Completion task: create a supervision rubric and project milestone plan.